

Lab 5: Memory Request Scheduling

Introduction

Task 1: Getting Your Hands on Ramulator 2

Question 1: Baseline System Configuration

DRAM Architecture & Organization

- Standard: DDR4
- Organization Preset: DDR4_8Gb_x8 (8 Gigabit density per chip, with an 8-bit data interface).
- Channels: 1 channel.
- Ranks: 1 rank per channel.

Timing & Performance

- Timing Preset: DDR4_2400R (Operating at 2400 MT/s).
- Clock Ratio: 3 (The memory system operates at a frequency relative to the simulation base clock). Memory Controller Logic
- Controller Implementation: Generic
- Scheduling Policy: FRFCFS (First-Ready, First-Come-First-Serve), which prioritizes requests that hit an open row (row-buffer hits) and then older requests.
- Refresh Management: AllBank (Refreshes all banks simultaneously).

System Integration

- Address Mapping: RoBaRaCoCh (Row-Bank-Rank-Column-Channel). This mapping scheme determines how physical addresses are translated into DRAM coordinates, prioritizing row locality.
- Memory System Implementation: GenericDRAM.
- Frontend Connection: It is connected to a SimpleO3 (Out-of-Order) CPU model with a 64KB Last Level Cache (LLC).

Question 2

- The frontend's YAML configuration accepts a list of traces. Each trace in the list corresponds to a CPU core. The SimpleO3 frontend simulation length is determined by the number of retired instructions, not a fixed number of CPU cycles.
- The simulation terminates once all cores have retired the specified number of instructions.

Question 3

I ran one multicore simulation (2C L-Trace, 2C H-Trace) and two single core simulations (once 1C L-Trace, once 1C H-Trace) for 300K instructions per simulation. The number of CPU cycles per core can be gathered from the `cycles_recorded_core_x` statistic printed.

$IPC = \text{num_expected_insts} / \text{cycles_recorded_core_X}$

Trace Type	Single-core IPC	Multi-core IPC	Diff
High Intensity (H)	0.602	0.416	-31%
Low Intensity (L)	3.22	2.174	-32%

Citations